

ZoneCast

User Manual

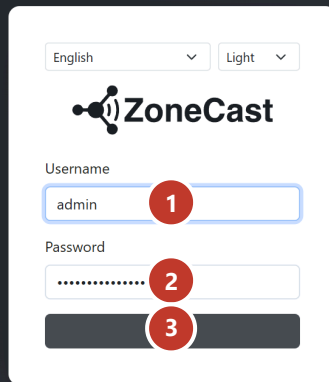
Illustrated guide to the dashboard's features

Application version 1.5.6 — September 2026

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1. Login



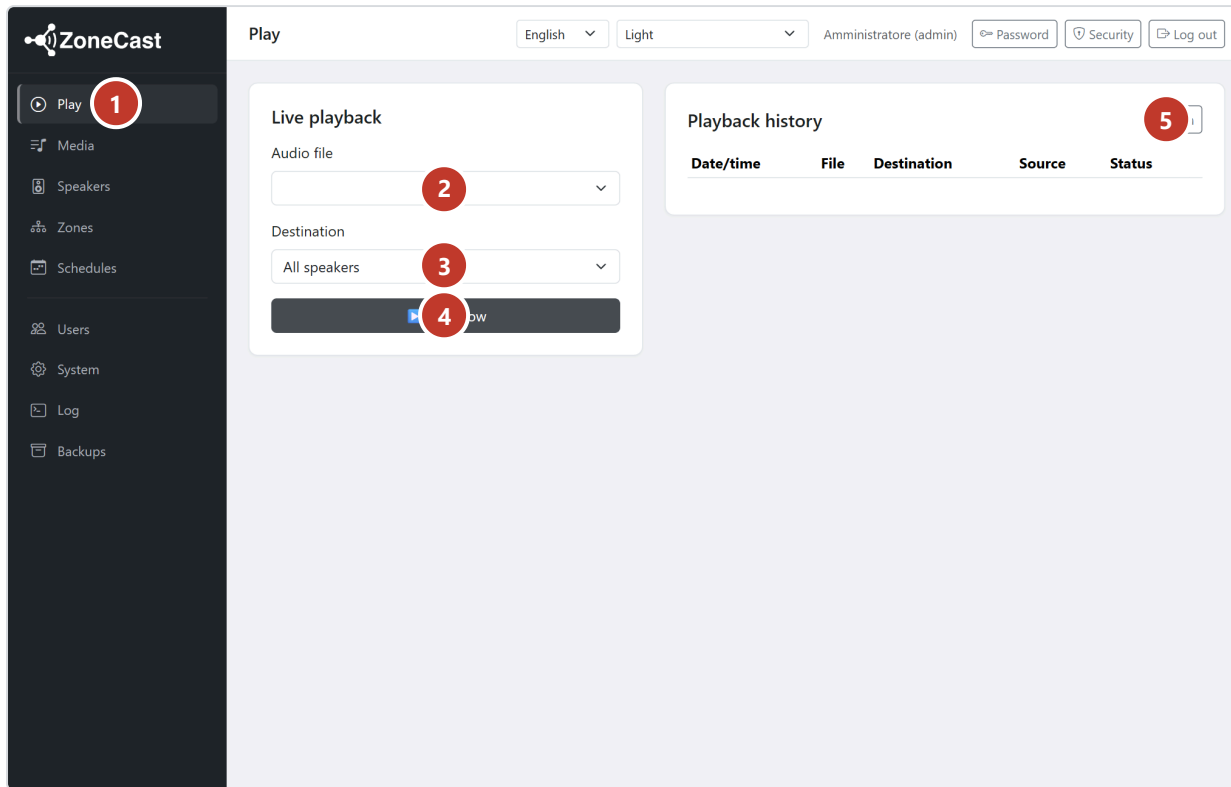
The login page is reachable at the server's address (e.g. `http://192.168.187.10`).

1. **Username** field — the username assigned by the administrator.
2. **Password** field.
3. **Sign in** button — if the account has 2FA enabled, after submitting you're also asked for the 6-digit authenticator app code (or a recovery code).

The login form also offers a **language** and **theme** (light/dark/automatic) picker at the top — see section 10.

After 5 failed attempts within 5 minutes, login for that user is temporarily blocked (429 — Too many attempts).

2. Live playback



This is the landing screen: it lets you start an immediate announcement to one or more speakers.

1. Sidebar menu — switch between dashboard sections.
2. **Audio file** — choose among the files already uploaded in Media.
3. **Destination** — a single speaker, a zone, or "All speakers".
4. **Play now** button — starts multicast playback immediately.
5. **Playback history** — outcome (completed/stopped/failed) of every manual or scheduled playback.

3. Speakers — list and actions

The screenshot displays the ZoneCast web interface for managing speakers. The left sidebar contains navigation links: Play, Media, Speakers (selected), Zones, Schedules, Users, System, Log, and Backups. The main content area is titled 'Speakers' and includes a header with language and theme selectors, user information, and security links. Below the header is a table of registered speakers. The table has columns for Status, Name, IP, Zone, Multicast group, and Model. Three speakers are listed, each with a set of action buttons. Red circles with numbers 1 through 7 highlight specific UI elements: 1. '+ New speaker' button, 2. Status dot, 3. Multicast group, 4. Ping button, 5. Preview button, 6. Apply now button, and 7. Backup / Edit / Delete buttons.

Status	Name	IP	Zone	Multicast group	Model	Actions
Offline	Altoparlante 1	192.168.187.52	—	239.1.3.2:5004	fanvil A233	Ping Preview Apply now Backup Edit Delete
Online	Altoparlante 2	192.168.187.53	—	239.1.3.2:5004	fanvil A233	Ping Preview Apply now Backup Edit Delete
Offline	Altoparlante 3	192.168.187.54	—	239.1.3.3:5004	fanvil A233	Ping Preview Apply now Backup Edit Delete

List of all registered speakers, with real-time online/offline status.

1. **+ New speaker** — opens the creation form (see section 4).
2. **Status dot** — green online, red offline.
3. **Multicast group** assigned to that speaker (dedicated address:port).
4. **Ping** — checks the device's reachability right away.
5. **Preview** — shows what would be sent to the device without actually applying it.
6. **Apply now** — writes the multicast configuration to the device (supported brands only).
7. **Backup / Edit / Delete** — under each row: back up the device's configuration, edit its data, delete it (backups remain available in the archive even after deletion).

4. Adding/editing a speaker

The screenshot shows the ZoneCast web interface with the 'Speakers' section active. A modal window titled 'Speaker' is open, allowing the addition or editing of a speaker. The form contains the following fields, each marked with a red circle and a number:

- 1. Name:** A text input field containing 'Entrance Speaker'.
- 2. IP address:** A text input field containing '192.168.187.60'.
- 3. Zone:** A dropdown menu with '(nessuna)' selected.
- 4. Dedicated multicast group:** A text input field containing '239.1.2.15'.
- 5. Brand:** A dropdown menu with 'Fanvil' selected.
- 6. Web admin user:** A text input field containing 'admin'.

Other fields in the form include:

- Model:** A text input field containing 'A233'.
- Port:** A text input field containing '5004'.
- Paging volume:** A dropdown menu with 'Not managed by ZoneCast (leave unchanged on device)' selected.
- Location:** A text input field containing 'Main entrance'.
- Web admin password:** A text input field with the placeholder 'leave empty to keep unchar'.

A 'Save' button is located at the bottom right of the modal. In the background, a table lists existing speakers with columns for Status, Name, and a list of actions (Ping, Preview, Apply now, Backup, Edit, Delete).

- 1. Name** — the label shown everywhere in the dashboard.
- 2. IP address** of the device on the LAN.
- 3. Zone** — optional grouping for per-area playback.
- 4. Dedicated multicast group** — address/port used to reach this single speaker.
- 5. Brand** — dropdown with the most common multicast brands (Fanvil, Algo, CyberData, Grandstream, Axis, Tiptel, Akuvox); choosing **Other** lets you type a name freely. The Apply now/Backup buttons and the announcement volume field only appear for brands with supported automatic actions.
- 6. Device web credentials** — used by the app to apply configurations; the password is encrypted at rest in the database.

5. Zones

ZoneCast

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English Light Amministratore (admin) Password Security Log out

Zones

1

Name	Description	Multicast group	# Speakers	
Capannone 1		239.1.2.1:5004	0	EditDelete
Capannone 2		239.1.2.2:5004	0	EditDelete
Capannone 3		239.1.2.3:5004	0	EditDelete

Zones group multiple speakers together (e.g. a floor, a department) so you can reach them together from Playback or from Schedules.

- + New zone** — name, description and zone multicast group.
- # Speakers** column — how many devices belong to this zone.

Sort any column with a clickable header (A→Z / Z→A arrows) in the Speakers, Zones, Media and Schedules tables.

6. Media — upload and audio analysis

The screenshot shows the ZoneCast web interface. On the left is a dark sidebar with navigation links: Play, Media (selected), Speakers, Zones, Schedules, Users, System, Log, and Backups. The main content area is titled 'Media' and includes a language dropdown (English), a theme dropdown (Light), and user information (Amministratore (admin)) with links for Password, Security, and Log out.

Upload audio file
Supported formats: MP3, WAV, M4A/AAC, OGG, FLAC, WMA, OPUS, AIFF — max 50 MB, max duration 10.0 minutes.

Below the upload section is a table of audio files. The first row, 'siren1.wav', has a red badge '1' on the upload button, a red badge '2' on the audio analysis badge, and a red badge '3' on the download button. The other two rows, 'clock1.wav' and 'beep1.wav', have no such badges.

File	Duration	Size	Uploaded on	Audio analysis	Actions
siren1.wav	57.0s	4.79 MB	9/23/2026, 3:52:57 PM	B 11% M 6% H 93%	Download Delete
clock1.wav	11.5s	0.97 MB	9/23/2026, 3:38:01 PM	B 1% M 6% H 93%	Download Delete
beep1.wav	7.8s	0.66 MB	9/23/2026, 3:37:49 PM	B 0% M 7% H 93%	Download Delete

Every uploaded file is automatically analyzed and assessed for playback on PA horns (which reproduce low frequencies poorly).

- 1. Upload** button — any audio format, automatically converted for playback.
- 2. Audio analysis** badge — a warning icon and color (green/yellow) if bass is dominant or significant relative to mid/treble: hovering shows a tooltip detailing the frequency breakdown. If an **"Amplify +N dB"** button also appears, the file has headroom to be brought up to maximum volume without distortion (one click applies the gain).
- 3. Download / Delete** the file.

The low/mid warning badge signals that the audio might sound weak or unclear on PA horns: it doesn't prevent playback, it's just a hint.

7. Schedules and holidays

The screenshot shows the ZoneCast web interface. On the left is a dark sidebar with the ZoneCast logo and a list of navigation items: Play, Media, Speakers, Zones, Schedules (highlighted), Users, System, Log, and Backups. The main content area is titled 'Schedules' and includes a header bar with language ('English'), theme ('Light'), and user ('Amministratore (admin)') dropdowns, along with buttons for Password, Security, and Log out. Below the header is a 'Schedules' table with columns: Name, File, Destination, Time, Days, Holidays, and Active. A '+ New schedule' button is in the top right of the table area. The table currently shows 'No schedules'.

Name	File	Destination	Time	Days	Holidays	Active
No schedules						

Plan automatic playback of a file at recurring times and days.

1. Every schedule has: media file, destination (speaker/zone/all), time, days of the week.
2. **Holiday calendar** — each schedule picks a reference country (Italy, United States, United Kingdom, France, Germany, Spain, Japan, China) and then whether to exclude that calendar's holidays, always include them, or trigger the announcement *only* on holidays.
3. **Active/Paused** switch to temporarily disable without deleting.

8. Users

The screenshot shows the ZoneCast web interface. On the left is a dark sidebar with navigation links: Play, Media, Speakers, Zones, Schedules, Users (highlighted), System, Log, and Backups. The main content area is titled 'Users' and includes a header with language ('English'), theme ('Light'), and user status ('Amministratore (admin)') dropdowns, along with 'Password', 'Security', and 'Log out' buttons. Below the header is a table of users with columns 'User', 'Name', and 'Role'. The table lists two users: 'admin' (marked as 'default') with role 'admin', and 'manutenzione' with role 'operator'. Each row has an 'Edit' button, and the 'manutenzione' row also has a 'Delete' button. A '+ New user' button is in the top right of the table area.

User	Name	Role
admin default	Amministratore	admin
manutenzione	manutenzione	operator

Only administrators can see and manage this section.

1. Available roles: **admin** (full access, including the Users/System/Logs/Backup sections) and **operator** (playback, managing speakers/zones/media/schedules).
2. The admin user created on first startup can't be deleted (always guarantees a way back in).
3. Every user can change their own password and enable 2FA on their own (see section 13) — an administrator can also reset another user's password from here.

9. System — admin panel

ZoneCast

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System

English Light Amministratore (admin) Password Security Log out

1

Network

Interface
ens160

IP address / subnet (CIDR)
192.168.187.20/24

Gateway
192.168.187.1

DNS servers (one or more, comma-separated)
192.168.187.1, 1.1.1.1

Apply

Must be confirmed within a few minutes from the dashboard reachable at the new address, otherwise it reverts on its own.

2

Set time

Local time
2026-09-24 12:45:12 UTC

Timezone
Etc/UTC

NTP sync status
Synced

NTP servers configured on the host
time.google.com, time.windows.com

NTP sync active

System timezone
Etc/UTC

Apply

NTP servers
time.google.com, time.windows.com

Apply

One or more, separated by space or comma. Empty = system default servers.

Set time manually
gg/mm/aaaa --:--:--

Set

Changes the VM's clock — if NTP is active, it's automatically disabled to keep the value set (otherwise it would be resynced right after).

3

Host resources

Disk
5.2 / 9.7 GiB (53.2%)

RAM
0.5 / 1.6 GiB (30.8%)

CPU (1 cores)
33.3%

4

Export full configuration

Full backup of database, media and encryption key in a single file. Contains sensitive data: handle it with care.

Backup password (optional)

Export and download

Information

Theme and language can be changed from the top bar, for any user.

Version
v1.5.6

Import configuration

Restores a previously exported backup. **Replaces** the current data (a copy is still saved) and restarts the service.

Scegli file Nessun file selezionato

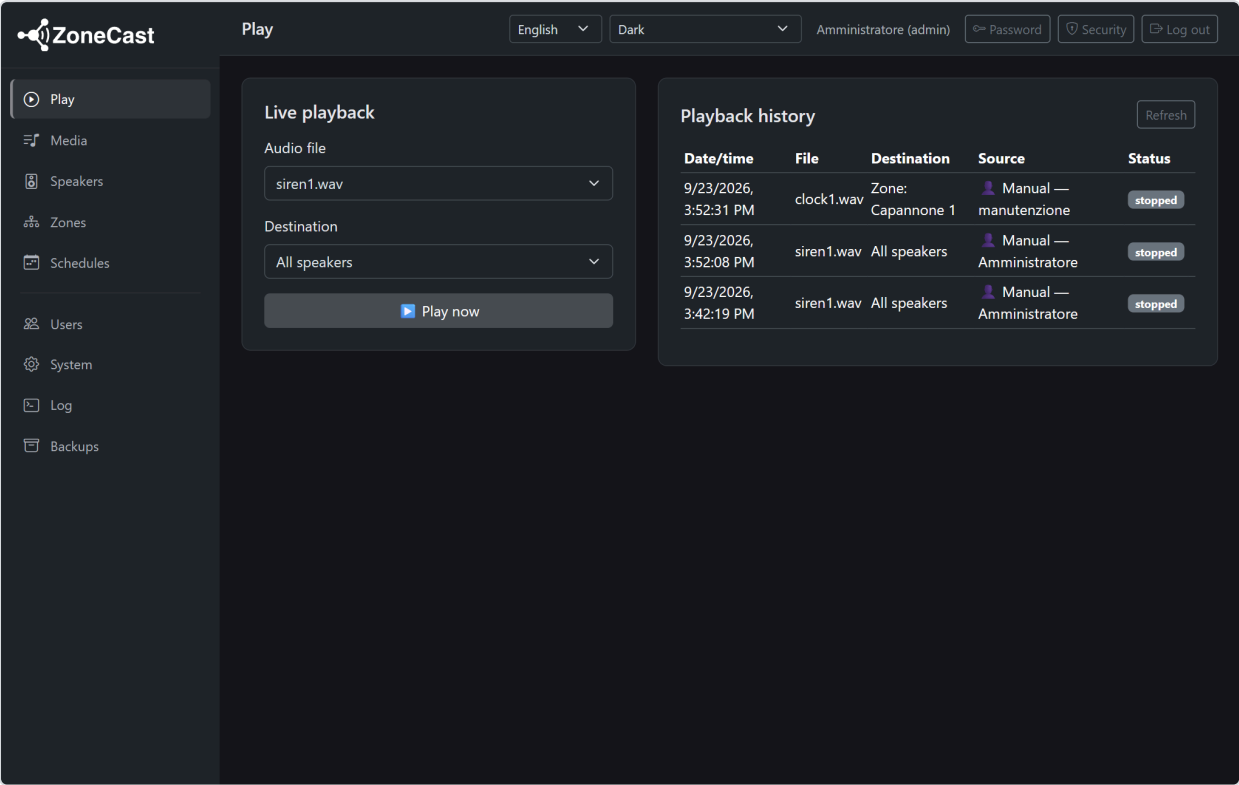
Backup password (if set during export)

Import and restart

Visible only to administrators.

- 1. Network** — IP address, subnet, gateway, DNS; changes are applied with a trial period (the interface shows the remaining time and must be confirmed from the new address, otherwise it automatically reverts to the previous configuration).
- 2. Server time** — NTP sync status, configurable NTP servers, timezone, or manual setting if the server has no NTP access.
- 3. Host resources** — disk usage (with a color-coded bar warning when space is running low), RAM and CPU.
- 4. Appearance and information, Export, Import** — see section 10 for appearance/language/version; further down, **Export full configuration** (database, media, device backups, encryption key) into a single password-protected file, to transfer the entire installation to a new machine, and the matching **Import** function to restore it.

10. Language, theme and version/changelog



Selectors available to **every** user (not just admins), always visible at the top of the dashboard and on the login page; on small screens they're found in the pop-out sidebar menu.

- 1. **Language** — English, Italian, Chinese, Japanese, German, Spanish, French: translates the whole interface, including the audio-analysis tooltips.
- 2. **Theme** — light, dark, or **automatic** (default): switches from light to dark following the real local sunrise/sunset time.

The language and theme choice stays saved in the browser: each user can have their own preference independently of others.

ZoneCast

System

Changelog

- v1.5.6** 2026-09-24
 - Fixed the playback history's source (Manual/Scheduled) and zone/all-speakers destination labels always showing in Italian regardless of the selected interface language
- v1.5.5** 2026-09-24
 - Fixed the 2FA status text in the Security modal always showing in Italian regardless of the selected interface language
- v1.5.4** 2026-09-24
 - Fixed the playback history table only showing the time of each entry, not the date, making older entries ambiguous
- v1.5.3** 2026-09-24
 - New schedules no longer default to excluding Italian public holidays — holiday filtering is now off by default and fully opt-in per schedule, for any of the supported countries
- v1.5.2** 2026-09-23
 - Operators can now create/edit/delete zones and speakers (including moving a speaker between zones), matching what the dashboard already let them attempt — only device actions (Apply now, Backup) and the admin-only sections stay admin-restricted
- v1.5.1** 2026-09-23
 - Fixed a crash when an operator (non-admin) account saved a schedule, caused by re-removing already-hidden admin-only nav items on every refresh
- v1.5.0** 2026-09-23
 - Fixed NTP server display showing the resolved IP address instead of the configured hostname
 - Language and theme selectors moved into the mobile sidebar drawer, decluttering the top bar on small screens

Host resources

Disk

RAM

CPU (1 cores)

Export full configuration

Full backup of database and sensitive data: handle it carefully

Backup password

Restart

The **Information** card in the System panel shows the installed application's version number and the list of changes per release (changelog), useful for quickly checking what changed after an update.

On the first login of the day, if there were recently failed schedules or disk space is running low, an automatic summary alert appears.

11. Event log

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Event log

All levels Search message... Refresh Export CSV Export JSON Clear log

Keep history for 30 days ☒ Never (no automatic deletion) Save

Date/time	Level	Source	Message
9/24/2026, 12:44:26 PM	INFO	zonecast.auth	Login riuscito: utente 'admin' da 10.81.234.130
9/24/2026, 12:43:46 PM	INFO	zonecast.scheduler	Loaded 0 active schedules
9/24/2026, 12:40:42 PM	INFO	zonecast.auth	Login riuscito: utente 'admin' da 10.81.234.130
9/24/2026, 12:39:12 PM	INFO	zonecast.auth	Login riuscito: utente 'admin' da 10.81.234.130
9/24/2026, 12:39:00 PM	INFO	zonecast.scheduler	Loaded 0 active schedules
9/24/2026, 12:34:15 PM	INFO	zonecast.auth	Login riuscito: utente 'admin' da 10.81.234.130
9/24/2026, 12:33:47 PM	INFO	zonecast.auth	Login riuscito: utente 'admin' da 10.81.234.130
9/24/2026, 12:07:16 PM	WARNING	zonecast.auth	Login fallito per utente 'admin' da 10.81.234.130
9/24/2026, 12:07:08 PM	INFO	zonecast.scheduler	Loaded 0 active schedules
9/24/2026, 9:56:25 AM	INFO	zonecast.scheduler	Loaded 0 active schedules
9/24/2026, 9:54:24 AM	INFO	zonecast.auth	Login riuscito: utente 'admin' da 10.81.234.2
9/24/2026, 9:51:09 AM	WARNING	zonecast.auth	Login fallito per utente 'admin' da 10.81.234.2
9/24/2026, 9:51:04 AM	INFO	zonecast.scheduler	Loaded 0 active schedules
9/24/2026, 6:12:36 AM	INFO	zonecast.speakers	Altoparlante 'Altoparlante 2' (192.168.187.53): offline -> online
9/24/2026, 6:12:36 AM	INFO	zonecast.speakers	Altoparlante 'Altoparlante 1' (192.168.187.52): offline -> online
9/24/2026, 6:07:40 AM	INFO	zonecast.speakers	Altoparlante 'Altoparlante 2' (192.168.187.53): online -> offline

Persistent record of logins, device communication errors, playbacks and schedules — survives a service restart. Admins only.

1. Filter by **level** (Info/Warning/Error) or search the message text.
2. **Export CSV/JSON** — downloads the full history (not limited to the current view) for archiving or sharing.

12. Backup archive

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Backups

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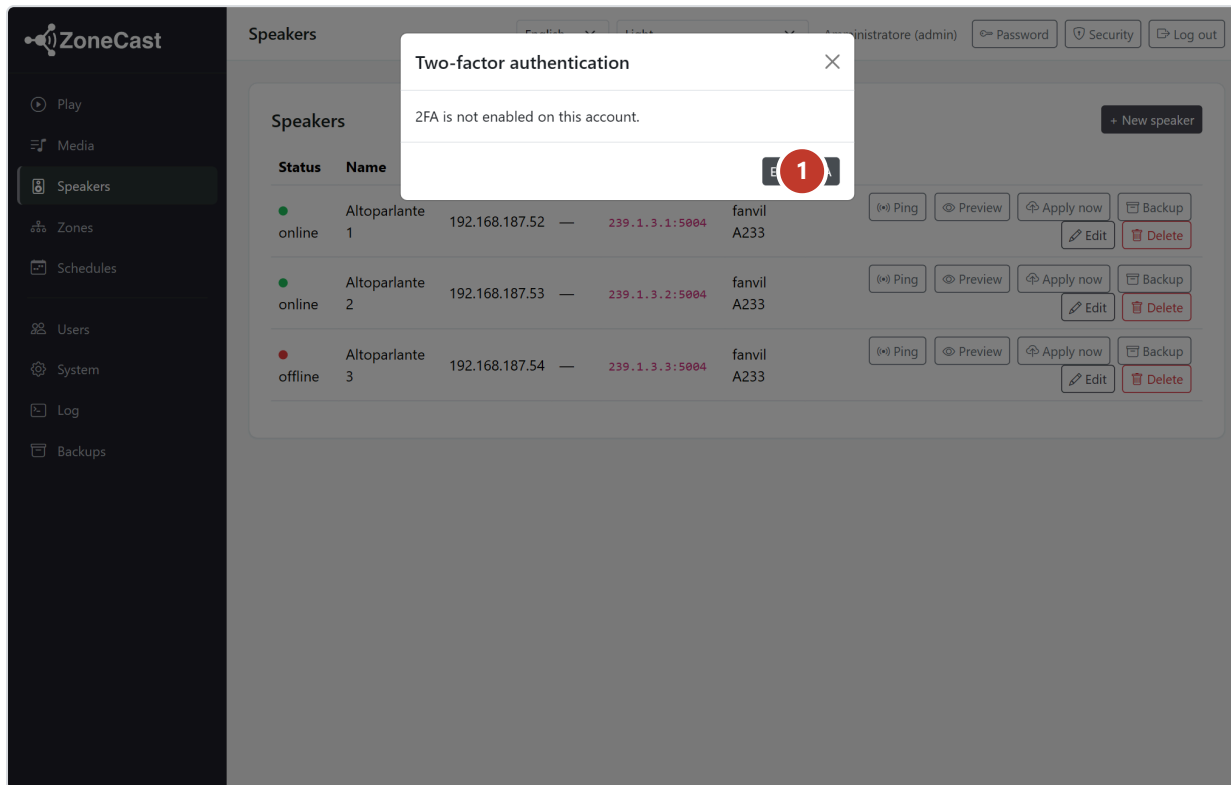
Backup archive

Date	Speakers	IP	Format	Size	Created by		
9/23/2026, 3:43:35 PM	Altoparlante 2	192.168.187.53	txt	37.0 KB	Amministratore	Download	Discard
9/23/2026, 3:43:33 PM	Altoparlante 1	192.168.187.52	txt	37.1 KB	Amministratore	Download	Discard

All device configuration backups, including ones whose speakers have since been deleted from the app.

1. Each row shows the speaker's name/IP *at the time of the backup*, not its current one — so it stays identifiable even if the device was removed or reassigned.
2. Download or permanently delete a backup from the archive.

13. Account security — 2FA



Accessible to any user from the **Security** button at the top, regardless of role — 2FA is optional, not enforced by the administrator.

- 1. Enable 2FA** — shows a QR code to scan with an authenticator app (Google Authenticator, Authy, etc.); after confirming with the first 6-digit code, 10 single-use recovery codes are generated, shown only once.
- 2.** From here you can also disable 2FA or regenerate the recovery codes, confirming with your password.

Keep the recovery codes somewhere safe: they're the only way in if you lose the phone with the authenticator app.